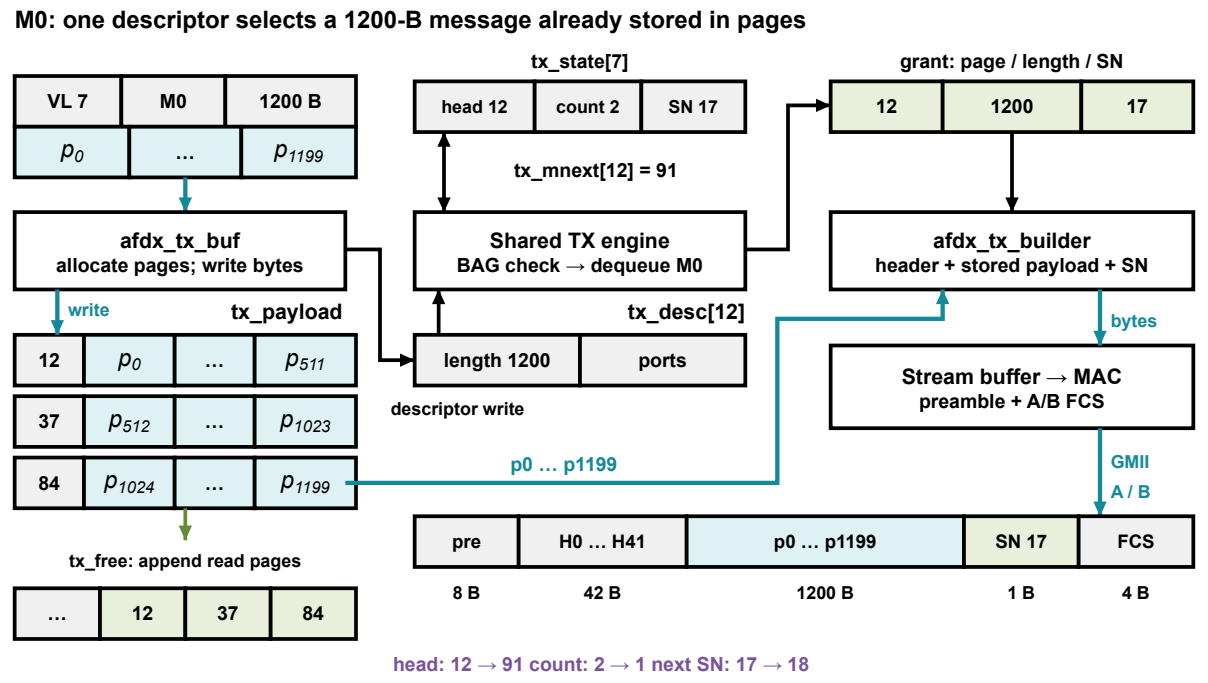
M0: one descriptor selects a 1200-B message already stored in pages
VL 7
M0
1200 B
p_0
…
p_{1199}
afdx_tx_buf
allocate pages; write bytes
write
tx_payload
12
p_0
…
p_{511}
37
p_{512}
…
p_{1023}
84
p_{1024}
…
p_{1199}
tx_free: append read pages
…
12
37
84
tx_state[7]
head 12
count 2
SN 17
tx_mnext[12] = 91
Shared TX engine
BAG check → dequeue M0
tx_desc[12]
length 1200
ports
descriptor write
grant: page / length / SN
12
1200
17
afdx_tx_builder
header + stored payload + SN
bytes
Stream buffer → MAC
preamble + A/B FCS
GMII
A / B
p0 … p1199
pre
H0 … H41
p0 … p1199
SN 17
FCS
8 B
42 B
1200 B
1 B
4 B
head: 12 → 91 count: 2 → 1 next SN: 17 → 18